02 · Lesson spec

Mini-project: Robot Says with Ayo · 40 minutes (can run over 2 sessions) · teams of 3 · fully unplugged · station check starts

Goal	**I can program cardboard Ayo with three picture cards and fix a wrong one.**
Core parts	Code + Build
Key words	Review: program · first · next · last · fix
Materials	Cardboard Ayo and a taped straight path per team; box targets (school gate or market stall signs); forward/stop/walk/wave/↻ cards; build card and picture build checklist (Printables); Station 1 kit: routine cards, ↻ cards, Station 1 record sheet (Guide 05)



1. Launch (5 min)

- **Say:** “You are coders now! Your team will **program** cardboard Ayo to reach a target. What does Ayo need from us?” (three cards in the right order, and a check for mistakes)
- Show the **build card** (6 steps) and the picture build checklist: order ✓ act ✓ check ✓ fix ✓.
- Give roles: **card-layer**, **robot**, **checker**. Each team chooses a target: school gate or market stall.

2. Build (25 min)

- **Build (Core):** stand Ayo at the start of the path; put the box target at the end.
- **Code (Core):** the card-layer lays a 3-card program (e.g. forward, forward, stop). The team acts it as friend-robots, then pushes Ayo one step per card.
- A partner from another team **swaps two cards**; the team finds and swaps them back, then runs it again.
- Teams tick their picture build checklist as they go. The teacher **photos** each finished program.
- **Stretch:** 4 cards, a ↻ card, or a second target.

Meanwhile: Station 1 (teacher, station check starts)

- Take **one small group at a time** (3–4 children, about 4 minutes) to Station 1. Each child: ① orders 3 routine cards → ② says first, next, last → ③ does one move 3 times with ↻ → ④ finds and swaps back two cards you swapped.
- Record E / D / S / X on the Station 1 record sheet (Guide 05). Aim for **8–12 children** this period. Continue in Week 12.
- Teams keep building while you assess: checkers keep them on track; an assistant helps if available.

3. Share (10 min)

- **Create:** each team presents: “Ayo goes… first ___, next ___, last ___!” and runs the program.
- If possible, **show another class** (now or in Week 13).
- **Close:** “You planned, built, tested and fixed. That is what engineers and coders do!”

Differentiation

Group	Build	Teacher role
Support	2-box path; 3 cards with the teacher laying the first	Sit with them between Station 1 groups
Core	3-card program to a box target; fix two swapped cards	Check-ins and photos
Stretch	4 cards, a ↻ card, or two targets	Challenge cards; celebrate at the Week 13 showcase

Common mistakes

- Pushing Ayo before the program is finished → “Plan first! All cards down.”
- Two pushes for one card → “One card, one step.”
- Reshuffling everything after a swap → “Check each card; find the two.”

What if it doesn’t work?

- Not enough cardboard Ayos → any small toy or a box with a face drawn on it.
- You cannot leave teams to run Station 1 → run the build as a whole class today and Station 1 in Week 12 only (all children must be seen by Week 13 morning).

How we check

Picture build checklist (not marked) + teacher observation + photo. Station 1 record sheet for the station check.

> **Safety**
> Light, empty box targets; no climbing; adults cut any cardboard; teams spread out on separate paths.